Bacterial–Nanostructure Interactions

A Literature Review

Biophysics · MSc in Engineering Physics · Instituto Superior Técnico

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Abstract: The rise of antimicrobial resistance (AMR) has driven the investigation of mechano-bactericidal nanostructured surfaces as alternatives to chemical agents. This report reviews the state of the art in bacterial–nanostructure interactions, covering the theoretical mechanics of cell wall deformation, the experimental methodologies employed to quantify adhesion and rupture forces, and the predictive biophysical models that describe membrane failure. Critical limitations of current static elastic models are identified, and future research directions toward clinical translation are proposed.

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1 Introduction and State of the Art

1.1 Clinical Motivation

The emergence of antimicrobial resistance (AMR) constitutes a critical global public health burden, currently attributed to approximately 700,000 deaths annually [1]. Conventional chemical strategies — antibiotics, disinfectants, and biocidal coatings — are progressively rendered ineffective by the capacity of bacterial populations to evolve resistance phenotypes and to shelter within protective *biofilms* [2]. A biofilm is a structured community of bacteria encased in a self-produced extracellular polymeric substance (EPS) matrix. The matrix impedes antibiotic penetration and suppresses the host immune response, rendering sessile cells orders of magnitude more tolerant to chemical agents than their planktonic counterparts [3].

The clinical impact is particularly acute in orthopaedic implantology. Infection rates in total hip arthroplasty (THA) and total knee arthroplasty (TKA) are reported at 1.23% and 1.21%, respectively, carrying a mortality rate of approximately 2.5% and an average episode cost of $50,000 [4]. Conventional titanium implant surfaces, characterised by micron-scale roughness, provide an adhesion-permissive substrate on which bacterial colonisation precedes osseointegration in the so-called *race for the surface* [4].

1.2 Towards Physical Mechano-Bactericidal Surfaces

In response to these limitations, research attention has shifted towards *mechano-bactericidal* surfaces, which induce cell lysis through purely physical mechanisms. Unlike chemical strategies, physical rupture does not engage the metabolic pathways through which resistance is acquired; the mechanism acts independently of bacterial genotype [2]. Surface topography at the nanoscale — rather than chemical composition — is identified as the primary determinant of bactericidal efficacy (BE) [5].

A critical distinction must be drawn between two classes of antibacterial surface action that are frequently conflated in the literature. *Anti-adhesive* surfaces reduce bacterial docking through electrostatic repulsion, steric exclusion (e.g., PEG brushes), or superhydrophobicity; they do not kill cells, and bacteria that overcome the energy barrier remain viable. *Bactericidal* surfaces, by contrast, induce irreversible membrane disruption and cell death upon contact [2]. This report is concerned exclusively with the latter mechanism. The distinction is not merely semantic: anti-adhesive performance is typically assessed by adhesion counts, whereas bactericidal performance requires viability assays, and conflation of the two metrics has produced inconsistent claims of efficacy in the literature.

1.3 Natural Archetypes

Nature provides well-characterised archetypes for this approach. The wings of *Psaltoda claripennis* (cicada) were demonstrated by Ivanova et al. to support arrays of nanopillars (height $\approx$ 200–300 nm, tip diameter $\approx$ 60 nm, centre-to-centre spacing $\approx$ 170 nm) capable of lysing *Pseudomonas aeruginosa* upon contact without chemical intervention [2]. Dragonfly wing nanostructures (*Diplacodes bipunctata*) exhibit a less-ordered but comparably effective topography [2]. More recent work on *Psaltoda linearis* cicada wings reports a similar pillar geometry with comparable bactericidal action, extending the natural-archetype evidence base [2]. Gram-negative bacteria were found to be more susceptible than Gram-positive species under identical topographic conditions, a differential attributed to the relative thickness of the peptidoglycan layer (1–7 nm vs. 20–100 nm) [2]. Notably, this trend is not universal across Gram-positive phenotypes: *Bacillus cereus* has been reported to exhibit higher susceptibility than *Staphylococcus aureus* despite both being Gram-positive, attributed to differences in cell morphology (rod vs. coccus) and local peptidoglycan cross-linking density [2].

These biological observations motivated the fabrication of synthetic analogues on materials relevant to clinical use, including black silicon (bSi), titanium dioxide (TiO_2), and gold nanostructures. Table 1 summarises the principal natural and synthetic topographies discussed in the literature.

Table 1: Representative natural and synthetic mechano-bactericidal topographies.

Source	Type	H (nm)	D (nm)	Material
Cicada wing	Pillars (ordered)	200–300	40–100	Chitin
Dragonfly wing	Random features	200–300	40–100	Chitin
Black silicon	Nanoneedles	400–800	20–50	Si
Anodised Ti	Nanotubes	—	60–70	TiO_2
E-beam Ti	Nanorough	$\sim$500	—	TiO_2

1.4 Scope of This Report

This report evaluates: (i) the mechanical properties of the bacterial cell wall and the physical principles governing membrane deformation; (ii) the contact mechanics and adhesion models used to predict rupture; (iii) the experimental methodologies employed to measure sub-microscopic forces and elasticity; and (iv) the limitations of current models and the research directions required for clinical translation.

2 Motivation

2.1 Bacterium

A bacterium is a unicellular organism. It comprises of a cell body, flagellum and pili (Fig. 1, [6]). A vital component of the bacterium is cytoplasm, which contains water, enzymes and other cellular structures. Genetic material is not enclosed within a nucleus - it is dispersed throughout the cytoplasm. Cytoplasm is enclosed via the inner membrane. Its primary function is protection, regulating the influx and efflux of materials. It is highly selective. The inner membrane is built with phospholipids and proteins. Next, the bacterium contains a cell wall. This rigid structure is comprised of peptidoglycan. Bacteria are divided into two groups regarding the thickness of the cell wall and the existence of an outer membrane. Gram-negative bacteria have a single-layered cell wall (1 - 7 nm). Gram-positive bacteria have multiple cross-linked layers (20 - 100 nm). To compensate for the thinness of the cell wall, gram-negative bacteria contain also an outer membrane (Fig. 2). The cell wall exhibits elasticity and flexibility, used to maintain the mechanical integrity of the cell. Various mechanical stresses include the turgor pressure, which is a stress induces by the cytoplasmic fluid. Outer membrane damage leads to compromising the cell wall, which can induce significant deformation of cell death [2].

Bacteria exhibit a strong affinity for surface attachment. They generally thrive within a biofilm environment. Biofilm is a community of bacteria in a self-produced extracellular matrix, also a source of infection. To prevent biofilm formation, numerous techniques incorporate chemical strategies. Such strategies include anti-adhesive principles, contact active principle or a biocide attached/release principle [3]. Physical strategies rely on mechanical properties. One of these physical techniques is based on nanopattern surfaces.

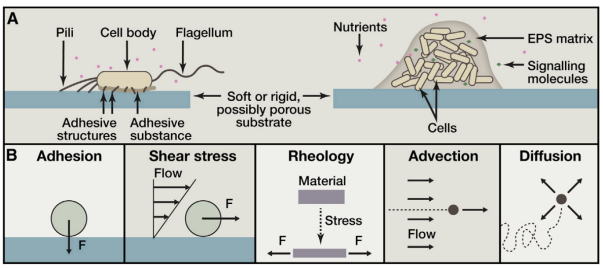

Figure 1: [6]

2.2 Nanostructures

Bacteria have shown a remarkable efficiency of defence against biochemical attacks. The adaptation is most visible especially in the case of antibiotics. A myriad of mechanisms that outsmart traditional biochemical approaches

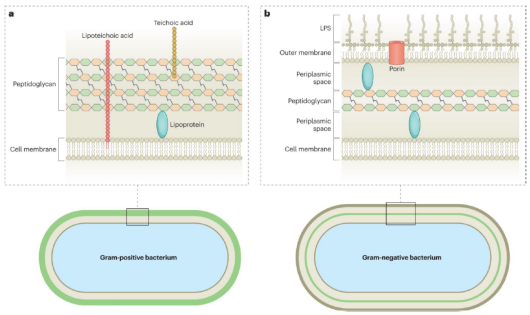

Figure 2: The structure of a bacterium. (*a*) Cell membrane and a peptidoglycan wall of a gram-positive bacterium. (*b*) Cell membrane, peptidoglycan wall and outer membrane of a gram-negative bacterium [2].

lead to the investigation of alternative techniques. One of those techniques is base on nanostructures. The idea of using nanostructures as an antibacterial techniques stems from its existence in nature. They are found on certain insect wings and animal skins. Ivanova et al. (2012) showed that nano-pattern on cicada wings can penetrate bacteria, leading to their death. Gram-negative bacteria exhibit susceptibility to the nano-patterns (causing their demise), whereas gram-positive bacteria are more resistant. Even the slightest changes in nano-pattern topology (such as pillar radius or pillar spacing) significantly impact its effectiveness [2]. Nanopattern surfaces are believed to cause bacteria-death by a surface attachment force called adhesion. The cell envelope of the uni-cellular organism gets destroyed via contact to the point of rupture. This process can be further studied in postmortem micrographs that show turgor loss (deflation) and cytoplasm leakage. Specific physical or mechanical parameters such as envelope stiffness, extensibility, pattern geometry and adhesion strength also aid the efficiency of the rupture of the bacterium [7].

3 Surface interactions

3.1 Adhesion

A bacterium first approaches surface via a fluid (motile bacteria use their appendages, non-motile depend on Brownian motion). Adherence to the surface is governed by the extended DLVO theory, which incorporates van der Walls forces, repulsive electrostatic forces and acid-base interaction (can be repulsive or attractive depending on the bacteria-substrate interaction, Fig. 3). The DVLO theory generates two energy minima separated by a significant energy layer. It predicts interpretative results especially for negative binding processes (attraction forces). However, for negatively charges surfaces (such at titanium or TiO_2 surfaces), the theory is not accurate, as it predicts a smaller barrier than reality. The first part of adhesion is reversible. The bacteria finds itself in the secondary energy-minima. To overcome the barrier, appendages are employed. However, this process is still reversible, as the appendages act as springs. The second part of adhesion is irreversible. The bonds strengthen as atoms reposition themselves. This irreversibility is further enhanced with the production of EPS (extracellular polymeric substance). As cells adhere to the surface, deformation occurs which amplifies the bonding process. Additionally, bacteria possess the remarkable ability of mechanosensoring, where the cell immediately responds to mechanical forces. The adhesion process suggests that bacteria create strong irreversible attachment to surfaces, which, when adhering to nanostructures, can be exploited [2].

3.2 Deformation

The second physical phenomena that is relevant when studying bacterium-surface interactions is deformation. In order to induce cell lysis, the cell membrane needs to be ruptured via deformation: an applied force whose effects depend on the material's elasticity. This property is measured in units of pressure (Pa) and describes how "soft" or "rigid" a material is. Softer materials, such as cells and bacteria have elastic modules in the range of 10kPa-100MPa, while stiffer materials such as bone and steel lie in the range of 1GPa-100GPa [4].

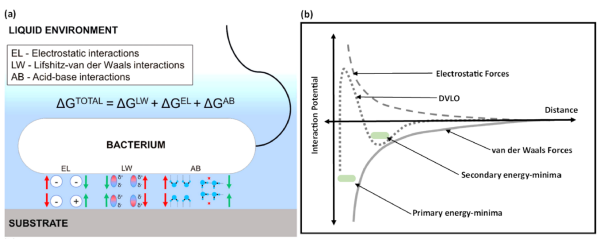

Figure 3: Adhesion forces. (*a*) Graphical illustration of extended DVLO (including acid-base interactions). (*b*) DVLO model: interaction potential dependence on distance. Secondary energy-minima and primary energy-minima represent reversible and irreversible attachment, respectively [2].

In this field, elasticity is often measured by Young's modulus (E), which is a type of elastic modulus. In the linear regime of the generalized form of Hooke's law, Young's modulus is the constant that relates stress (σ) and strain (ϵ), as shown in equation 1. It is an intrinsic property of the material, independent of its geometry.

$$\sigma = E\epsilon \tag{1}$$

The material's stiffness directly impacts the way it can be deformed. Therefore, thoroughly analysing and characterizing the elasticity of cells is central to building a realistic and effective physical model of cell rupture by nanoindentation. Three models widely used in current research in the field are the Hertz model, the Johnson-Kendall-Roberts (JKR) model and the Derjaguin-Muller-Toporov (DMT) model.

3.3 Physical models

3.3.1 Hertz model

The Hertz model is one of the most used models today for the characterization of elasticity of biological materials such as cells and bacteria. It was first proposed by Heinrich Hertz in 1882 [8] and opened up a whole new branch of physics for contact mechanics which further developed and studied the ideas set by Hertz. However, despite its age and new developments, the Hertz model remains relevant and applicable in many cases.

Later developed in a complete form by Kenneth Langstreth Johnson in his 1985 book *Contact Mechanics* [9], the Hertz model provides a simplified description of the interaction between two solid bodies in contact. It is restricted to frictionless surfaces and perfectly elastic materials, and does not account for adhesion, van der Waals forces, or chemical bonding effects. The model further assumes that the contacting surfaces are topographically smooth at the micro- and nanoscale. Under these conditions, the following assumptions are made:

1. The contact area between the two solids is elliptical
2. The solids are non-conforming bodies, such that stresses are localized near the contact region
3. The surfaces are frictionless
4. The strains are small
5. Each solid can be treated as an elastic half-space

This is a highly general theory applicable to any two bodies that satisfy these assumptions. In the context of microbiology, the Hertz model has been adapted to specific

experimental conditions, and numerous studies in the literature outline similar methodologies for data acquisition and fitting to the Hertz model [10, 11, 12].

Experimental data is obtained using Atomic Force Microscopy (AFM), which will be discussed in further detail in section 4. In summary, AFM operates by applying an external force to the sample via probing tip, inducing a deformation that is measured as a displacement in the direction of the applied force. The resulting data are represented as a force–indentation curve, typically interpreted within the framework of Hooke's law.

The goal is then to fit this experimental curve with the corresponding theoretical prediction from the Hertz model, allowing the extraction of the material's Young's modulus. In practice, this requires using specific force–indentation relationships, whose form depends on the geometry of the AFM tip:

$$F_{cone}(\Delta z) = \frac{2E^*}{\pi(tg(\alpha))}\Delta z^2 \tag{2}$$

$$F_{parabolic}(\Delta z) = \frac{4\sqrt{R}E^*}{3}\Delta z^{\frac{3}{2}} \tag{3}$$

$$\frac{1}{E^*} = \frac{1-\mu_{tip}^2}{E_{tip}} + \frac{1-\mu_{sample}^2}{E_{sample}} \tag{4}$$

In these expressions, E is Young's modulus, Δz is the indentation along the direction of deformation, z, and μ is Poisson's ratio. For a conical tip, α is the opening angle and for a parabolic tip, R is the radius of curvature at the apex.

The review by Anja Vinckier and Giorgio Semenza [10] provide a detailed step-by-step experimental guide on how to construct the force-indentation curves. The first step is to probe indentation on a hard surface, such as a glass substrate, as a reference. Then, without changing the configuration of the microscope, replace the hard reference surface by the soft sample and probe its indentation in the same way. The difference between the two curves, plotted in the same graph as shown in Figure 4, is the actual deformation of the soft sample. In these experiments, the indentation of the tip is seen as negligible.

3.3.2 Johnson-Kendall-Roberts (JKR) model

Hertz theory accurately predicts the elastic behaviour of solids under high loads. However, the model strongly deviates from experimental data for loads close to zero, as is stated in K. L. Johnson *et al.* [13]. According to the authors, at low loads, the contributions of attractive forces (van der Waals) become non-negligible. Thus, the Johnson-Kendall-Roberts (JKR) model was developed as an expansion to the Hertz model that accounts for adhesion.

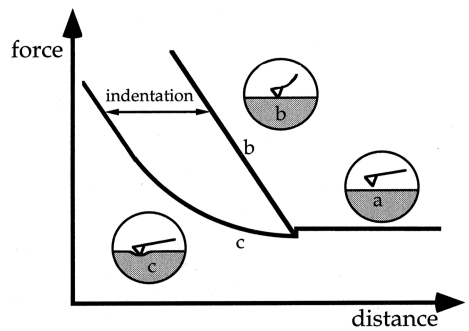

Figure 4: Schematic AFM graph for (a): no indentation, (b): indentation on a hard surface and (c): indentation on a soft sample [10].

Similarly to the Hertz model, the JKR model was first introduced in a simplified framework upon which new research developed. Today, works in the literature give ready-to-use expressions for simulations [14] and explore the effects of different geometries [15] (as the original model was derived for spherical contact only).

The relationship between force and indentation in the JKR model is given by:

$$F_{JKR} = \frac{4E^*a^3}{3R^*} - 4\sqrt{\pi\gamma E^* a^3} \tag{5}$$

Where a is the radius of contact area, γ is the interface energy, R^* is the effective contact radius and E^* still follows from equation 4. The first term is exactly the Hertz force, while the second term is the correction from adhesion added in the JKR model.

3.3.3 Derjaguin-Muller-Toporov (DMT) model

A second model that derives from Hertz and also includes adhesion is the Derjaguin-Muller-Toporov (DMT) model. It is not used as often in the research of bacterium-surface interactions as it mostly models stiff materials, whereas cells and bacteria have lower elastic moduli, and so are considered soft materials.

In the study by Ralph Spolenak *et al.*, the authors use Tabor's parameter μ to quantify the transition from JKR to DMT. In equation (6), γ is the work of adhesion and ϵ is the interatomic distance at the closest contact point.

$$\mu = \frac{R^{\frac{1}{3}}\gamma^{\frac{2}{3}}}{(E^*)^{\frac{2}{3}}\epsilon} \tag{6}$$

In Figure 5, a map is built for modelling biological attachment as opposed to modelling artificial attachment systems based on the parameters E^*, R and μ. As is shown in this estimation, biological systems will mostly not be well fit by the DMT model.

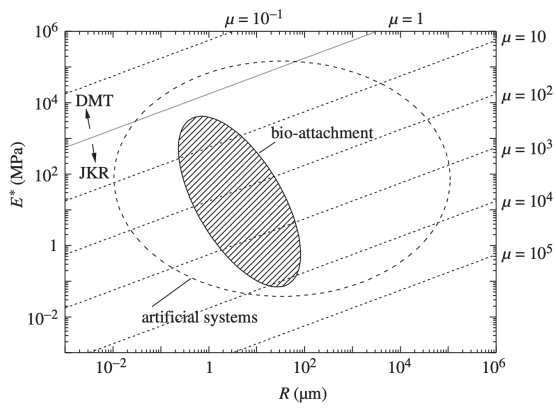

Figure 5: Comparison between the domain of JKR and DMT models [15].

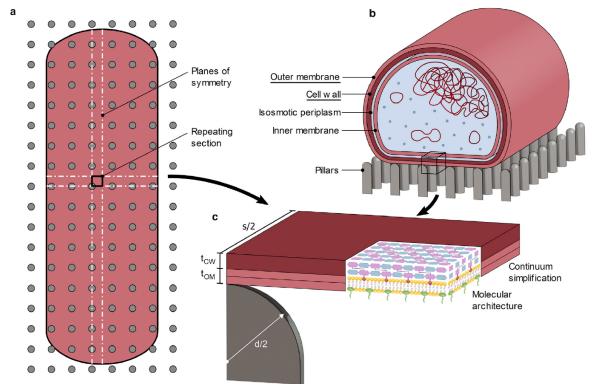

Figure 6: Configuration of the Velic model. (*a*) Geometric pattern of the nano-structure. (*b*) 3D cell on nanopillars. (*c*) Continuum simplification of molecular architecture [7].

3.4 Biophysical models

Due to the scale at which such physical interactions are happening, there is a huge attempt for biophysical modelling. Accurate biomechanical models are yet to be determined, as there are still many unknowns. Such encountered difficulties include a priori shape assumptions, monolayer envelope models or plane strain simplifications of a 3D problem, all of which lead to misrepresentative interpretations. The very first relevant biophysical model, which worked with nano-structured patterns to induce cell-death was conducted by Pogodin et al. (2013). The study proposes that the cell rupture occurs in a suspended region due to membrane stretching, without any penetration. However, it did not offer any reliable evidence to support the claim of rupture in the suspended region. Later, Xue et al. in 2015 used stretching theory in their research. They concluded that physical interactions are not sufficient alone to create complete adhesion and other forces are needed to induce cell rupture (such as van der Waals etc). Li et al. (2016) introduced additional bending energy and later (2019) discuss the effect of interfacial energy gradient. According to this model, when the pressure exceeds the critical elastic stress, cell wall undergoes creep deformation and ultimately penetration and death [7].

3.4.1 Velic et al. model (2025)

A recent study by Velic et al. (2025) worked on adhesion-driven envelope deformation model of a gram-negative bacterium. The aim to correctly identify the critical location of envelope deformation is key to understand the underlying mechanisms. A 3D model was employed, including multilayer structure. The nano-pillar geometry was determined to be a quare array of spherically capped, cylindrical pillars. This scheme mimicked the surface of the cicada wings. The curvature of the cell was negligible. A simplification of the cell was induced, as the only load-bearing layers incorporated in the process are the outer membrane and the cell wall. These layer were modelled using a 3D continuum approach. The outer membrane was modelled as an isotopic neo-Hookean material. Cell wall was modelled as a linear orthrotropic elastic material. Pillars were chosen as infinitely stiff (bending as cell periphery is not relevant for the surface-projected area where the rupture happens). The geometrical configuration with the 3D model is depicted in Fig. 6. Three different envelope configurations were modelled: soft, intermediate and stiff. Regarding contact relationships, there are three relative configurations. Surface adhesion is modelled as a closure with rough friction no separation. Membrane-leaflets interaction (the hydrophobic effect) is modelled via surface-to-surface contact with no friction, meaning the leaflets can slide off each other but cannot separate. Lastly, a tie constraint is induced to monitor the relative motion of inner membrane to the cell wall [7].

In Fig. 7 the three modelled curves are depicted with reference to actual data. Red zones are known area of rupture, whereas black curves indicate the model. When compared with micrographs, the soft and intermediate envelope correlate with the model. However, the stiff model does not enter the observed rupture-zone until 40 mJ/m^2, which is unrealistically high. The conclusion is, strictly adhesion-driven rupture can occur. However, the failure of the stiff envelope model can be attributed to the fact that the cited rupture strains were taken from experiments existing only for Gram-positive peptidoglycan. The peptidoglycan in Gram-positive cells is, however, much stronger due to cross-linking. Other discrepancies regarding the data and this model can be found in other ways of simplification, for example neglected stiffness of the outer membrane. Lastly, geometry of the nano-pillar model showed an interesting dependency, as shown in Fig. 8. A decrease in pillar radius leads to an increase in maximal strain. On the other hand, a change in spacing does not show any huge change regarding the adhesion work. That is in accord with experimental data [7]. Regarding the probe locations, in Fig. 9 an square element is studied. The critical location at each

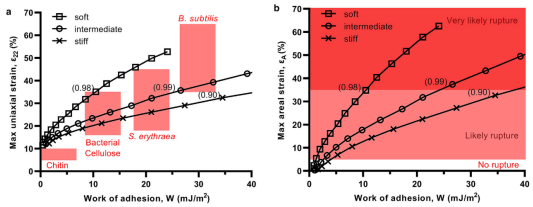

Figure 7: Estimated work of adhesion of three different Gram-negative envelopes. (*a*) Cell wall deformation by the maximal uniaxial strain in longitudinal direction (ε_{22}). (*b*) Outer membrane deformation by maximal areal strain (ε_A) [7].

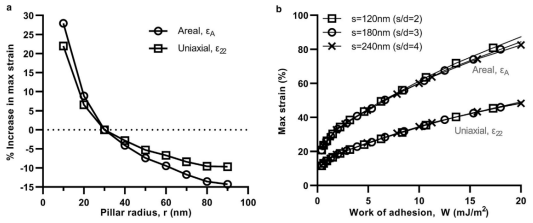

Figure 8: Adhesion correlated to nano-pillar geometry. (*a*) The dependence of increase in maximal strain on pillar radius. (*b*) The dependence of maximal strain for different nano-pillar spacing on adhesion work [7].

layer differs. Any stretch-induced rupture first initiates within the inner leaflet at the apex. The cell wall strains at a pillar apex were the largest. These findings contradict the dogma of a suspended region rupture.

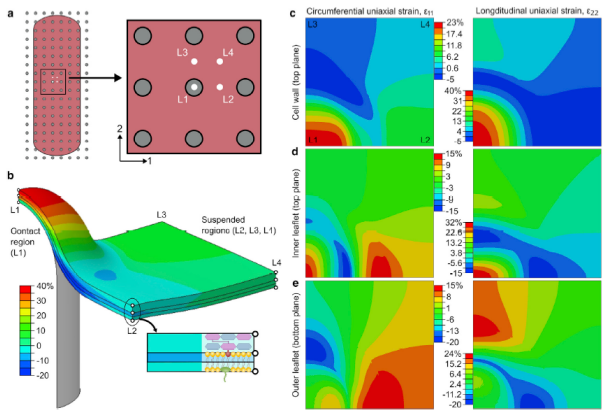

Figure 9: Probe locations. (*a*) The studied domain. (*b*) 3D geometry. (*c*) Unaxial strain at cell wall. (*d*) Unaxial strain at inner leaflet. (*e*) Unaxial strain at outer leaflet [7].

4 Experimental Methodologies

4.1 Atomic Force Microscopy and Nanoindentation

Atomic Force Microscopy (AFM) constitutes the principal instrument for quantifying the mechanical properties of individual bacterial cells at the nanoscale. In nanoindentation mode, a calibrated cantilever tip (spring constant typically 0.01–0.1 N/m) is brought into contact with the cell surface under controlled load, and the resulting force–displacement curve is recorded [4]. The Young's modulus (E) is extracted by fitting the contact regime of the force–displacement curve to an appropriate contact mechanics model. For a spherical indenter on a thin elastic shell (the bacterial approximation), the Hertz model gives:

$$F = \frac{4}{3}\frac{E^*}{\sqrt{R}}\delta^{3/2} \tag{7}$$

where F is the applied force, δ is the indentation depth, R is the effective tip radius, and $E^* = E/(1-\nu^2)$ is the reduced modulus (ν: Poisson's ratio) [4].

Reported Young's moduli span a wide range: Gram-negative *E. coli* has been measured at 0.1–50 MPa, while Gram-positive *S. aureus* values range from 10–200 MPa, reflecting the thicker cross-linked peptidoglycan layer [4]. This variability arises from hydration state, growth phase, applied load rate, and the choice of contact mechanics model.

A key limitation is that the Hertz model assumes a purely elastic, homogeneous, isotropic half-space, none of which conditions are satisfied by a bacterial cell. Application of Hertz contact to a thin-walled pressurised shell systematically overestimates E when indentation depth exceeds $\sim$10% of the shell wall thickness [4]. Additionally, the field of view of AFM is restricted to $\sim$100 μm^2, requiring careful selection of representative cells and precluding population-level statistics from a single experiment.

4.2 Single-Cell Force Spectroscopy (SCFS)

SCFS extends AFM to the direct measurement of adhesion forces between a single living bacterium and a target substrate [4]. A bacterial cell is immobilised on a tipless cantilever via poly-L-lysine coating or non-specific physisorption, and the cantilever is brought into contact with the nanostructured surface. The resulting retraction force–displacement curve yields the total adhesion force (F_{adh}, typically 0.1–10 nN) and the work of adhesion, which provides a direct experimental estimate of the thermodynamic parameter W in Eq. **??** [4].

SCFS measurements are intrinsically low-throughput: a single force curve requires 10–60 s, and typical experiments capture 30–100 cells per condition. Statistical uncertainty is therefore substantial, and reported values should be interpreted with regard to the coefficient of variation, which commonly exceeds 50% across a bacterial population [4]. Furthermore, SCFS does not capture the dynamic component of adhesion, as cells are held in static contact for a defined dwell time that may not reflect the transient collision events occurring in flow conditions.

4.3 Electron Microscopy

Scanning Electron Microscopy (SEM) and Transmission Electron Microscopy (TEM) are used to image the bacteria–nanostructure interface at nm resolution. SEM micrographs of bacteria on cicada wings and bSi nanoneedles consistently reveal deformation signatures including turgor loss (cell deflation), cytoplasm leakage, and membrane wrinkling at pillar contact zones [2]. Cross-sectional TEM imaging provides direct visualisation of pillar–membrane contact geometry, enabling measurement of actual indentation depth (δ) and the spatial relationship between the pillar apex and the zone of maximum deformation [7].

A critical observational limitation is that both SEM and TEM require fixation, dehydration, and/or staining protocols that alter cellular morphology. Turgor loss in particular cannot be attributed unambiguously to nanostructure-induced rupture versus the dehydration artefact inherent in sample preparation. Cryo-TEM, which preserves the hydrated state, is employed in select studies to mitigate this artefact, though throughput remains very low [2].

4.4 Fluorescence Microscopy and ROS Detection

Cell viability on nanostructured surfaces is quantified by fluorescence-based live/dead assays, most commonly using SYTO 9 (membrane-permeable; stains all cells green) and propidium iodide (PI; membrane-impermeable; penetrates only cells with compromised membranes, staining red) [2]. The ratio of PI-positive to total cells provides a direct measurement of the fraction with disrupted membranes, used as a proxy for bactericidal efficacy.

More recently, ROS-sensitive fluorescent probes (e.g., H_2DCFDA, CellROX) have been applied to detect reactive oxygen species (ROS) in bacteria exposed to nanostructured substrates [16]. Jenkins et al. demonstrated that nanopillar surfaces induce intracellular ROS accumulation concurrent with membrane disruption, suggesting that the mechanical stress imposed by the nanostructure triggers oxidative stress cascades in addition to direct physical rupture [16]. This finding introduces a mechano-chemical coupling dimension that is absent from purely mechanical models.

The principal limitation of fluorescence assays is the static nature of the experimental setup. Bacteria are typically deposited on surfaces under conditions of zero or

undefined fluid flow; the absence of physiological shear stress fundamentally alters both the adhesion probability and the force distribution on adhering cells [2].

4.5 Finite Element Analysis (FEA)

FEA complements experimental measurements by resolving the spatial distribution of stress and strain within the bacterial envelope at resolution inaccessible to direct measurement. The bacterial envelope is discretised as a continuum of finite elements with assigned constitutive laws — typically linear elastic or neo-Hookean for the membrane layers — and subjected to boundary conditions representing the nanostructure geometry and adhesion work [7].

The Velic et al. model employed a 3D multilayer continuum representation of a Gram-negative envelope, with the outer membrane modelled as an isotropic neo-Hookean material and the cell wall as a linear orthotropic elastic material [7]. The nanopillar array geometry was specified as a square-packed array of spherically capped cylindrical pillars of defined radius and height, mimicking the cicada wing surface. Siddiquie et al. extended this FEA framework to incorporate the viscoelastic (creep) response of the cell, addressing the time-dependent deformation of *E. coli* under sustained contact load [17].

A persistent limitation in FEA models is the reliance on constitutive parameters (modulus, rupture strain) derived from experiments on Gram-positive peptidoglycan, in the absence of equivalent data for the thinner Gram-negative cell wall [7]. This introduces systematic uncertainty into predicted rupture thresholds for the species most commonly studied in bactericidal surface experiments.

5 Discussion and Future Developments

5.1 Limitations of the Static Elastic Model

The dominant analytical framework in the bactericidal nanostructure literature treats the bacterial cell as a passive elastic sphere — a *pressurised elastic balloon* — subject to a prescribed work of adhesion. This approximation, embodied in the Pogodin et al. model [18] and its successors, provides useful order-of-magnitude estimates of rupture thresholds but systematically misrepresents several physical features of the living cell.

First, the bacterial envelope is not a linearly elastic solid. AFM creep measurements on *E. coli* demonstrate a time-dependent, power-law viscoelastic response under constant indentation load, with characteristic relaxation times in the range of seconds [17]. A purely elastic model applied to such data does not yield a physically meaningful Young's modulus; it reports an apparent stiffness that depends on the loading rate used in the AFM experiment. The consequence for rupture prediction is non-trivial: under the sustained contact that occurs when a bacterium adheres to a nanopillar, the viscoelastic membrane will deform progressively via creep beyond the elastic prediction, potentially reaching the rupture strain over a timescale of minutes rather than instantaneously [17].

Second, turgor pressure is not a fixed boundary condition. As the cell deforms, cytoplasmic volume changes alter P_t dynamically. Furthermore, mechanosensitive ion channels (e.g., MscL, MscS) open in response to membrane tension, allowing rapid ion efflux that reduces P_t within milliseconds [6]. A cell undergoing active turgor regulation will present a different force– deformation curve from a passive elastic shell, and this response is entirely absent from current FEA models.

5.2 The Stretching–Piercing Debate

Two mechanistic models have been proposed for the locus and mode of membrane failure on nanopillar arrays. The *stretching model*, formulated by Pogodin et al., predicts rupture in the *suspended* membrane region between adjacent pillars, driven by adhesion-induced lateral stretching [18]. The *piercing model*, supported by FEA results from Velic et al. and cross-sectional TEM observations, places the critical strain at the *pillar apex*, where the non-developable double curvature of the indented membrane imposes a localised stress concentration [7].

The Velic model demonstrates that the maximum uniaxial strain (ε_{22}) in the cell wall and the maximum areal strain (ε_A) in the outer membrane leaflet both occur at the pillar apex, contradicting the suspended-region rupture dogma [7]. The stiff-envelope case in that model fails to reach the rupture zone below $W = 40\,\mathrm{mJ\,m^{-2}}$, a value above the range reported for biological adhesion, suggesting that pillar-apex stress concentration — rather than global stretching — is the energetically accessible failure mode for realistic adhesion energies.

5.3 Mechano-Chemical Coupling and Oxidative Stress

The identification of ROS accumulation in bacteria on nanopillar surfaces by Jenkins et al. [16] indicates that the bactericidal mechanism is not exclusively mechanical. Membrane deformation below the rupture threshold may impose sufficient tension to activate ROS-generating membrane enzymes (e.g., NADPH oxidase homologues), initiating a lethal oxidative cascade. This mechano-chemical pathway implies that the effective bactericidal threshold is lower than the mechanical rupture strain alone, as sublethal mechanical stress may be amplified biochemically to a lethal outcome.

Current models that define cell death solely by a critical strain value therefore overestimate the mechanical force required for bactericidal action. A complete model of bacteria–nanostructure interaction must incorporate a coupled constitutive law relating membrane tension to ROS production rate, with a survival criterion based on both physical integrity and oxidative damage state.

5.4 The Dead-Cell Stagnation Problem

A practical limitation of static nanostructured surfaces is the accumulation of lysed bacterial debris on pillar tips following the initial bactericidal event. This debris, comprising membrane fragments, released cytoplasmic proteins, and EPS, effectively masks the sharp nanofeatures and creates a conditioning film that reduces the surface energy available for subsequent bacterial deformation [2]. Surfaces that demonstrate high initial BE in short-term assays (1–4 h) may therefore exhibit substantially lower efficacy in long-term or high-inoculum conditions. No quantitative model of this fouling dynamics has been reported to date, and the timescale of capacity depletion under physiological inoculum levels remains uncharacterised.

5.5 The Selectivity Problem

For orthopaedic implant applications, bactericidal activity must be achieved without compromising the adhesion and proliferation of mammalian osteoblasts (*race for the surface*) [4]. Osteoblasts are characterised by membrane surface areas three to four orders of magnitude greater than typical bacterial cells, and by a significantly higher membrane compliance. Under contact with an identical nanostructure, the localised stress experienced by an osteoblast membrane at a pillar tip is substantially lower

than that experienced by a bacterium, because the larger cell distributes the adhesion energy over a greater area [2]. This geometric selectivity is the physical basis for the observed preferential killing of bacteria while supporting osteoblast integration.

However, the selectivity is not universal. High-aspect-ratio nanofeatures with sub-50 nm tip diameters — effective against Gram-positive species — impose non-negligible stress on eukaryotic cell processes and filopodia. A rigorous selectivity criterion would require mapping the stress distribution in both cell types as a function of pillar geometry, a computation not yet reported in the literature.

5.6 Fluid Dynamics and Physiological Shear Conditions

In the majority of published viability assays, bacteria are deposited under static or near-static conditions, and rinsing protocols are performed at unspecified flow rates [2]. In clinical contexts, nanostructured implant surfaces are exposed to shear stresses ranging from $\sim$0.001 Pa (implant interstitium) to $\sim$37 Pa (arterial wall), depending on anatomical location. Fluid shear alters bacterial adhesion by modifying the frequency and duration of cell–surface contacts and by imposing drag forces that oppose the adhesion work. At physiological shear, only irreversibly adhered cells remain on the surface; the ratio of reversibly to irreversibly adhered cells is not captured in static assays.

The development of microfluidic platforms that impose controlled laminar or pulsatile shear while enabling fluorescence imaging of live/dead status represents a necessary methodological advance for translational validation.

5.7 Future Research Directions

The following research directions are identified as necessary to advance mechano-bactericidal surfaces towards clinical deployment:

Viscoelastic FEA models. Incorporation of a power-law creep constitutive law for the bacterial envelope, calibrated against AFM retardation spectra, would enable prediction of time-to-rupture under sustained adhesion and provide quantitative guidance on the minimum pillar height required for lethal deformation within a clinically relevant timescale.

Mechano-chemical coupling. Systematic co-measurement of membrane strain (via AFM nanoindentation) and ROS production (via H_2DCFDA fluorescence) on surfaces with defined pillar geometry would enable empirical construction of a strain–ROS transfer function, which could be incorporated as a second failure criterion in FEA models.

Standardised microfluidic testing. Adoption of a standardised flow-cell protocol, specifying shear stress (τ, in Pa), dwell time, and inoculum concentration, is required to generate comparable BE data across research groups and to support regulatory submissions for implant coatings.

Machine learning-assisted geometry optimisation. Supervised ML models trained on existing BE datasets have demonstrated 81% accuracy in predicting bactericidal outcome from topographic parameters [2], with water contact angle and feature height identified as the most predictive variables. Extension to include cell-wall mechanical parameters and shear conditions as input features could enable in-silico optimisation of nanostructure geometry prior to fabrication.

Smart/switchable surfaces. Stimuli-responsive polymer brushes (e.g., PNIPAM, thermoresponsive at $\sim$32°C) could be integrated with nanostructures to cyclically collapse and extend the brush layer, mechanically displacing lysed cell debris and regenerating the bactericidal surface over multiple killing cycles [2].

Conclusion

The physical interaction between bacteria and nanostructured surfaces is governed by a balance of adhesion energy, cell-wall elasticity, internal turgor pressure, and the localised stress concentration imposed by sub-100 nm feature geometry. The accumulated evidence from AFM nanoindentation, SCFS, cross-sectional TEM, and FEA modelling indicates that membrane failure initiates at the pillar apex rather than in the suspended inter-pillar region, contradicting the original stretching model of Pogodin et al. [18] and supporting the apex-rupture mechanism described by Velic et al. [7].

The identification of concurrent ROS accumulation in bacteria on nanopillar surfaces demonstrates that the bactericidal mechanism is not purely mechanical; sub-lethal mechanical strain appears to initiate oxidative stress cascades that contribute independently to cell death [16]. This mechano-chemical coupling lowers the effective rupture threshold below the value predicted by elastic models and must be incorporated into next-generation predictive frameworks.

Three principal barriers to clinical translation have been identified. The dead-cell stagnation problem — whereby lysed debris masks pillar tips and reduces long-term bactericidal capacity — lacks a quantitative model. The absence of standardised testing under physiological fluid shear conditions prevents direct comparison of BE across studies and limits confidence in *in vitro* predictions. The selectivity problem requires geometric optimisation that balances bacterial membrane rupture against osteoblast compatibility, a computation not yet performed rigorously.

Progress on these challenges requires the integration of viscoelastic constitutive models into FEA, the development of microfluidic validation platforms, and the empirical quantification of the strain–ROS relationship across bacterial phenotypes. Mechano-bactericidal nanostructures represent a physically sound and resistance- independent approach to infection control; realising their clinical potential depends on the resolution of these outstanding questions in quantitative cell mechanics.

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